Rumor Spreading: Fast Information Dissemination

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Executive Summary

We present an analysis of rumor spreading in a complete network using simple push/pull protocols.
We show that information disseminates in $\Theta(\ln n)$ rounds with high probability and outline the key ideas and experiments.
We also discuss variants (failures, multiple calls, influencers) and practical applications.

Motivation

- How quickly can a message reach all nodes using simple randomized rules?
- Examples: gossip updates, social media virality, etc.
- Surprisingly, trivial push/pull rules achieve dissemination pretty fast (in logarithmic time).

Model: Push/Pull

- Network: complete graph on n nodes, initially 1 node informed.
- **Push model:** Each informed node contacts one random neighbor per round.
- **Pull model:** Each uninformed node contacts one random neighbor per round.
- **Spread Time:** Measure the time T until all nodes are informed.
- **Communication Cost:** Measure the Total number of calls made through out the run of experiment.
- Communication cost is our time complexity of experiment.

Empirical Growth

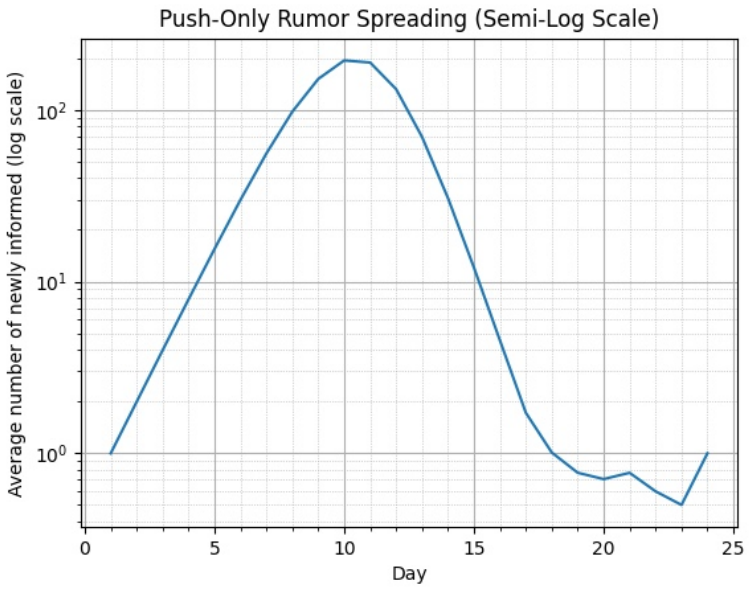


Figure: Newly informed people as rumor spreading proceeds

- Initially, the informed fraction grows rapidly (almost all calls are to an uninformed person).
- Growth slows down as more nodes become informed (many calls are redundant).

Core Intuition: Expected Growth

If $X_t = k$ nodes are informed, each of the $n - k$ uninformed avoids being contacted with prob. $(1 - 1/n)^k$.
Hence from Linearity of Expectation,

$$[X_{t+1} - X_t] = (n - k)\Big(1 - \Big(1 - \frac{1}{n}\Big)^k\Big).$$

For $k \ll n$, $(1 - \frac{1}{n})^k \approx e^{-k/n}$, giving $[X_{t+1}] \approx 2k$ (doubling of informed nodes).

Main Result

- Rumor spreads in $T = \Theta(\ln n)$ rounds *with high probability*.
- Lower bound (push): since informed nodes can at most double each round, $T \geq \log_2 n$ rounds.
- Conclusion: the randomized push protocol achieves near-optimal $\Theta(\log n)$ time.

Push vs Pull

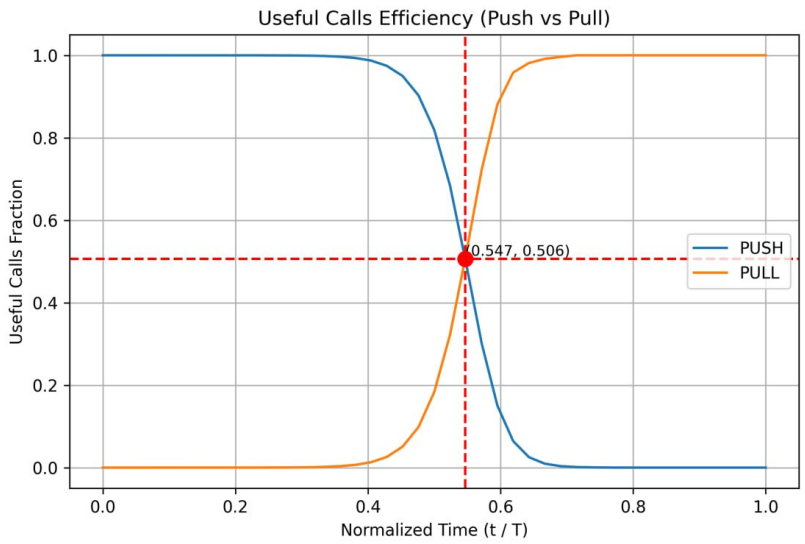


Figure: Comparison of PUSH, PULL, and Global Threshold

- **Push:** Very effective early on (few informed nodes contacting many uninformed).
- **Pull:** Gains traction later (uninformed nodes actively seek informed ones).
- Combined/Hybrid approach: use push in early rounds and switch to pull later for best of both.

Hybrid Threshold Approach

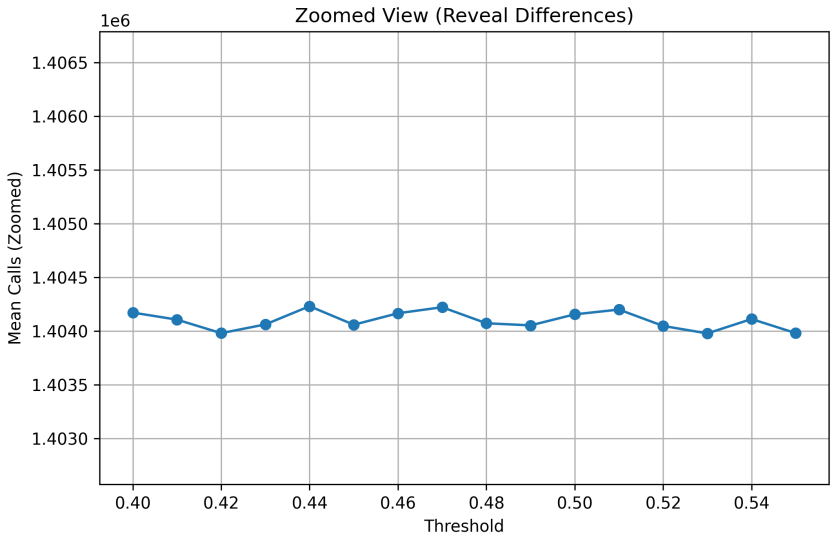


Figure: Comparison of Global Threshold

Limitation:

- Requires knowledge of global state ($\approx 50\%$ informed)
- Rapid changes within a round make precise switching difficult
- Performance becomes insensitive in range 0.5–0.6 (noisy plateau)

Conclusion:

- Global thresholding is impractical $\rightarrow$ need local, adaptive strategies

Hybrid Push-Pull (Karp et al.)

- Strategy: alternate or combine push and pull rounds adaptively.
- Achieves $O(\ln n)$ rounds and only $O(n \ln \ln n)$ total message transmissions.
- There is a matching $\Omega(n \ln \ln n)$ lower bound for any address-oblivious $O(\ln n)$-round protocol.

Hybrid Push-Pull (Karp et al.)

Idea: Nodes adapt behavior based on local state

States:

- **A (Uninformed):** Node does not know the rumor
- **B (Active spreading):** Node pushes & pulls rumor
- **C (Final spreading):** Node continues briefly to finish dissemination
- **D (Inactive):** Node stops participating

Transitions:

- A $\rightarrow$ B: when node first receives rumor
- B $\rightarrow$ B: continues spreading while growth is effective
- B $\rightarrow$ C: when most neighbors already informed (detected via counters)
- C $\rightarrow$ D: after $O(\log \log n)$ rounds

Hybrid Push-Pull (Karp et al.)

Counter Mechanism (Local Decision Rule):

- Each node in state B maintains a counter `ctr`
- Counter reflects local evidence of saturation

Update Rule:

- If a node receives rumor mostly from **higher counter nodes** $\rightarrow$ increment `ctr`
- If interactions are still with uninformed / lower counter nodes $\rightarrow$ keep `ctr` unchanged

Switching Condition:

- When $\texttt{ctr} \geq O(\log \log n) \rightarrow$ node switches to state C
- State C runs for $O(\log \log n)$ rounds (pull-dominated phase)
- Then node stops (state D)

Communication Cost

- Naive push (or pull) for $O(\ln n)$ rounds: $\approx n \ln n$ total messages.
- Karp's hybrid protocol: only $\Theta(n \ln \ln n)$ total messages.
- Trade-off: optimize time vs. number of calls; $n \ln \ln n$ is essentially optimal for $O(\ln n)$ time.

Variants

- **Failure probability (f):** Each call succeeds with prob. $1 - f$. Still $O\left(\frac{\log n}{1-f}\right)$ rounds if $f < 1$ is constant, but with larger constants.
- k **calls per node:** Each node contacts k neighbors per round. Speeds up: rounds $\approx \frac{\ln n}{\ln(1+k)}$.
- **Planted influencers:** Having m "influencer" nodes. Time $= O(\log\log n - \log m)$.
- **Influencer:** a node with high probability of getting called by another node

Concentration of Spreading Time

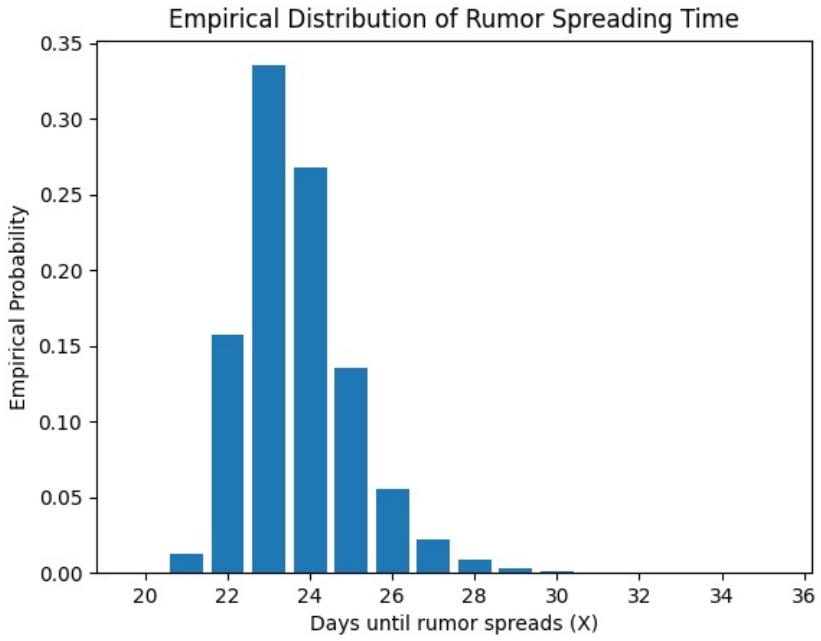


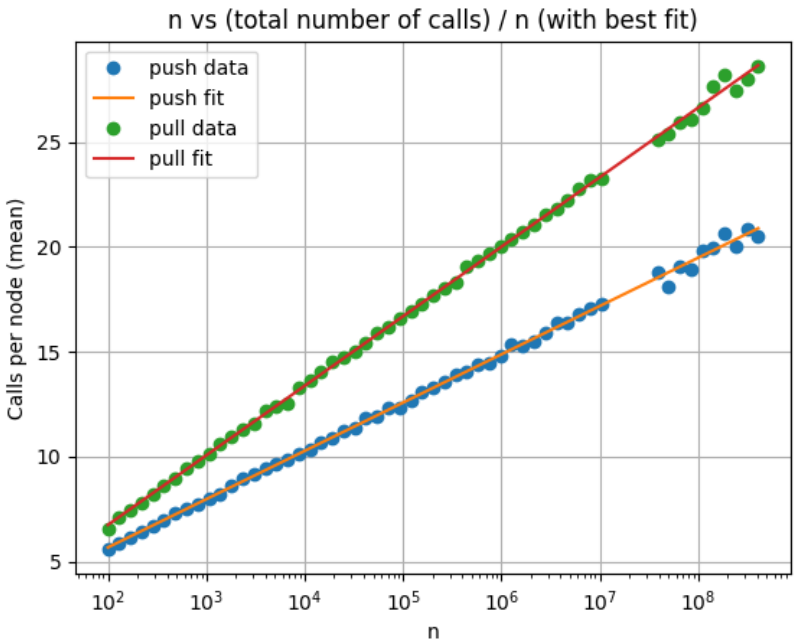
Figure: Probability distribution of spreading time

- The completion time T is concentrated around its expectation.
- Intuition: Changing one round or one call has limited effect (bounded differences).
- Experimentally: The bar-graph of T over trials is tight (small standard deviation).

Experimental Setup

- **Randomness:** Used RNG (xoshiro256++) with splitmix64 seeding for better performance.
- **Scaling of n:** Network size increased multiplicatively to efficiently capture asymptotic behavior.
- **Adaptive iteration:** More iterations for smaller n , and fewer for larger n, leveraging stronger concentration observed at scale.
- **Memory Optimization:** State stored using bit-level representation, significantly reducing memory footprint.
- **Metrics:** Dissemination time, fraction informed vs time, useful vs wasted calls.

PUSH vs PULL



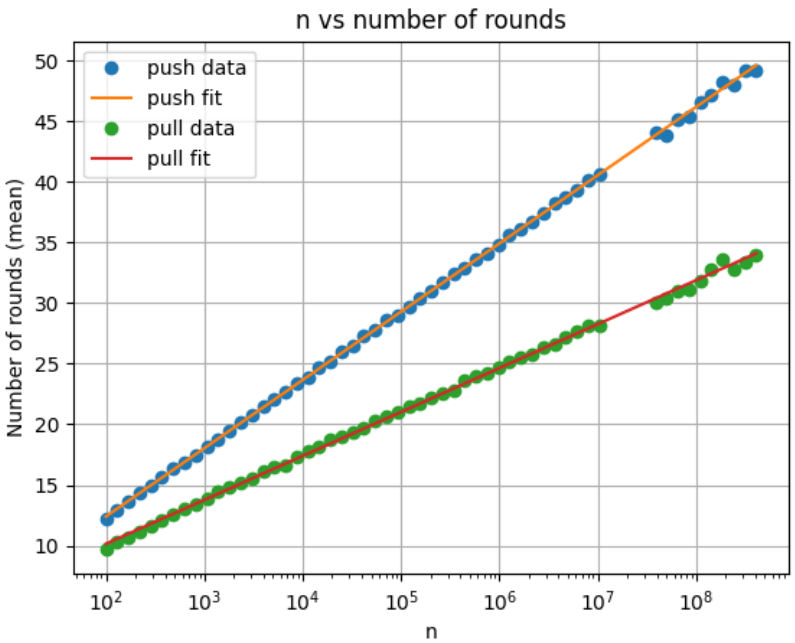


Figure: Left: Total number of calls vs log n. Right: Total number of rounds vs log n

- In case of minimizing total rounds PULL approach seems better.
- In case of minimizing total calls PUSH approach seems better.

Effect of Failure Probability

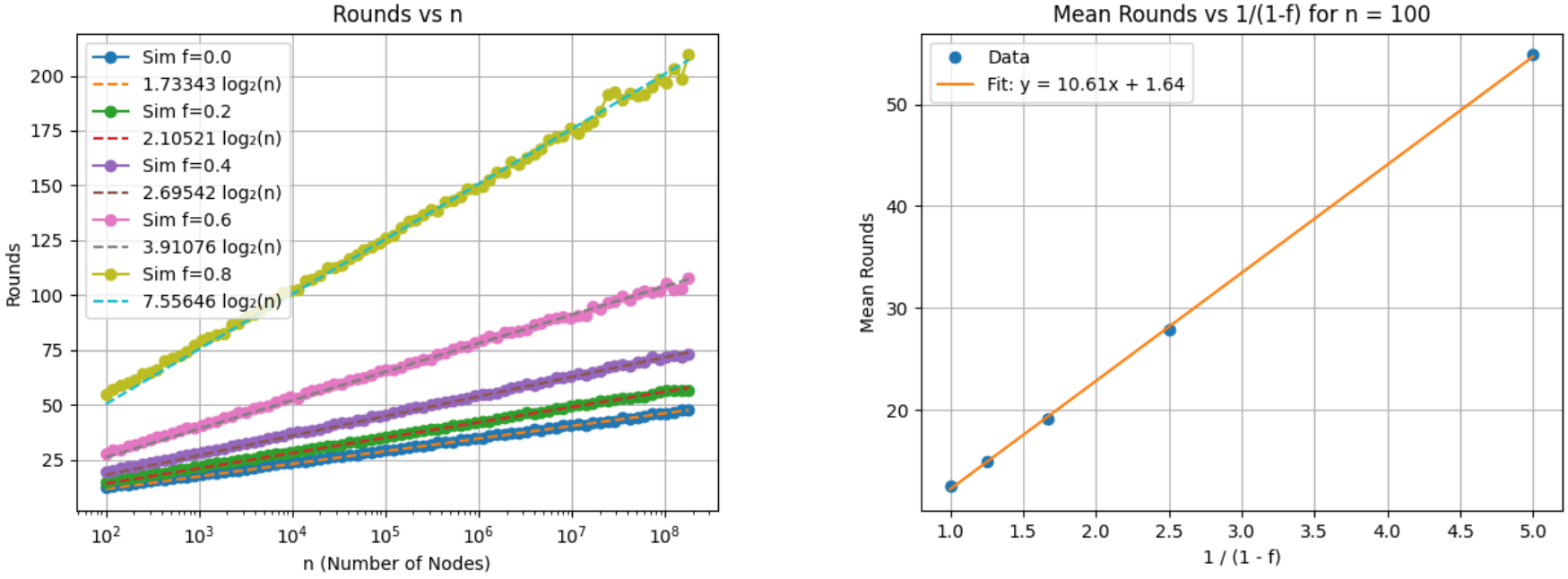


Figure: Left: Dissemination time variation for different failure probabilities f. Right: Dissemination time vs failure probability

- Increasing f slows down spreading
- Logarithmic trend persists for moderate f
- High f significantly increases constants

Multiple Calls per Node (k)

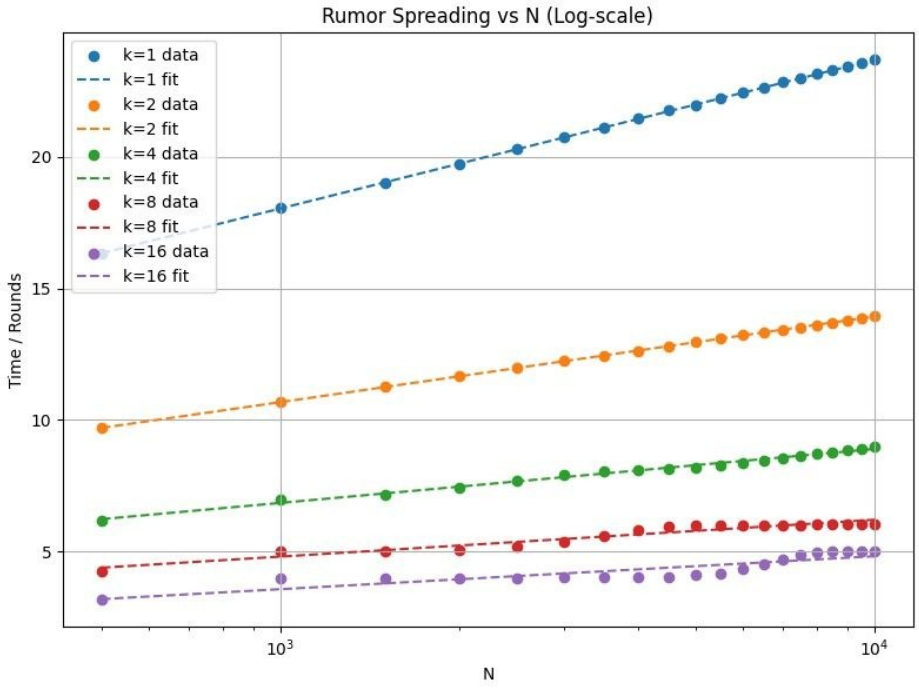
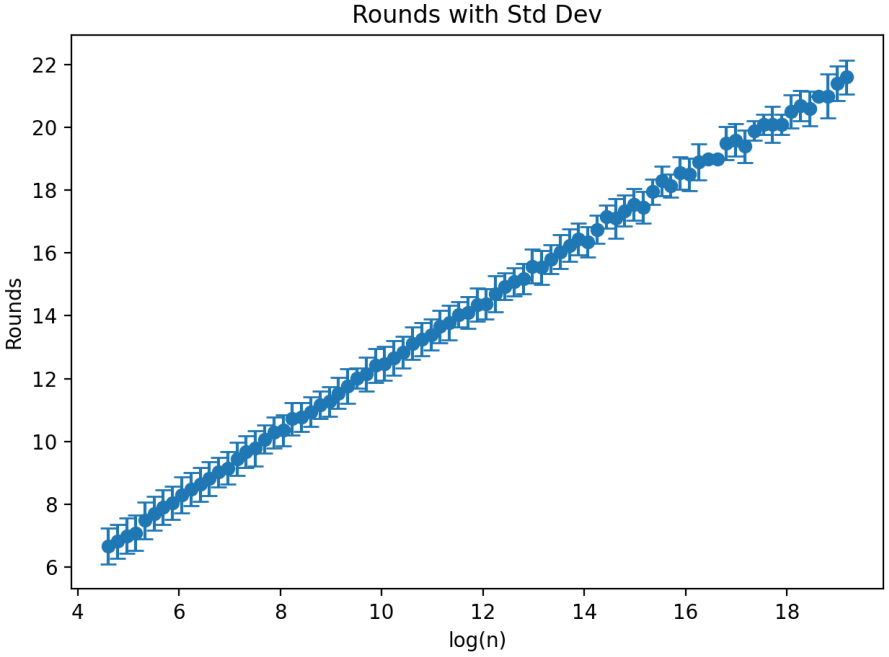


Figure: Rounds vs k

- Increasing k accelerates spreading
- Time reduces roughly as $\frac{\log n}{\log(1+k)}$
- Trade-off: higher communication cost

Karp et al. Algorithm



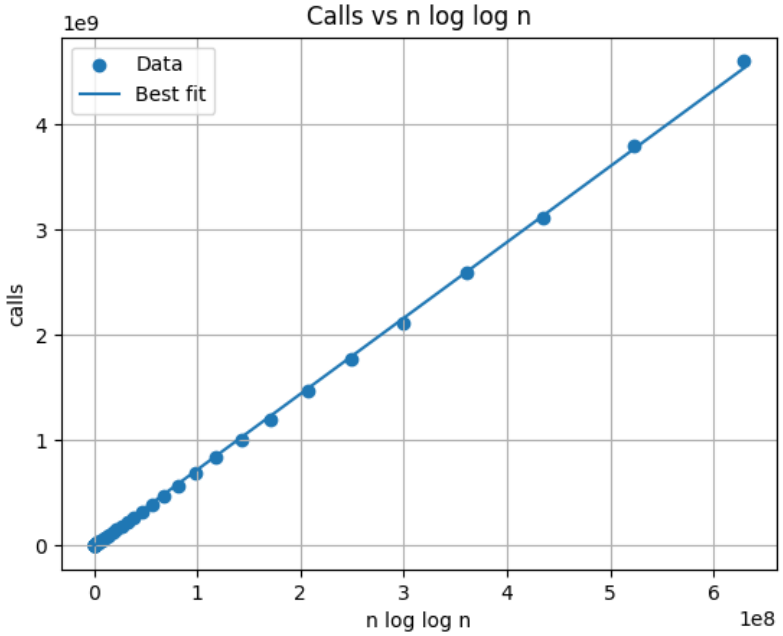


Figure: Time and calls comparison

- Maintains $O(\log n)$ time
- Reduces calls to $O(n \log \log n)$
- Adaptive phase detection avoids redundant calls

Planted Influencers

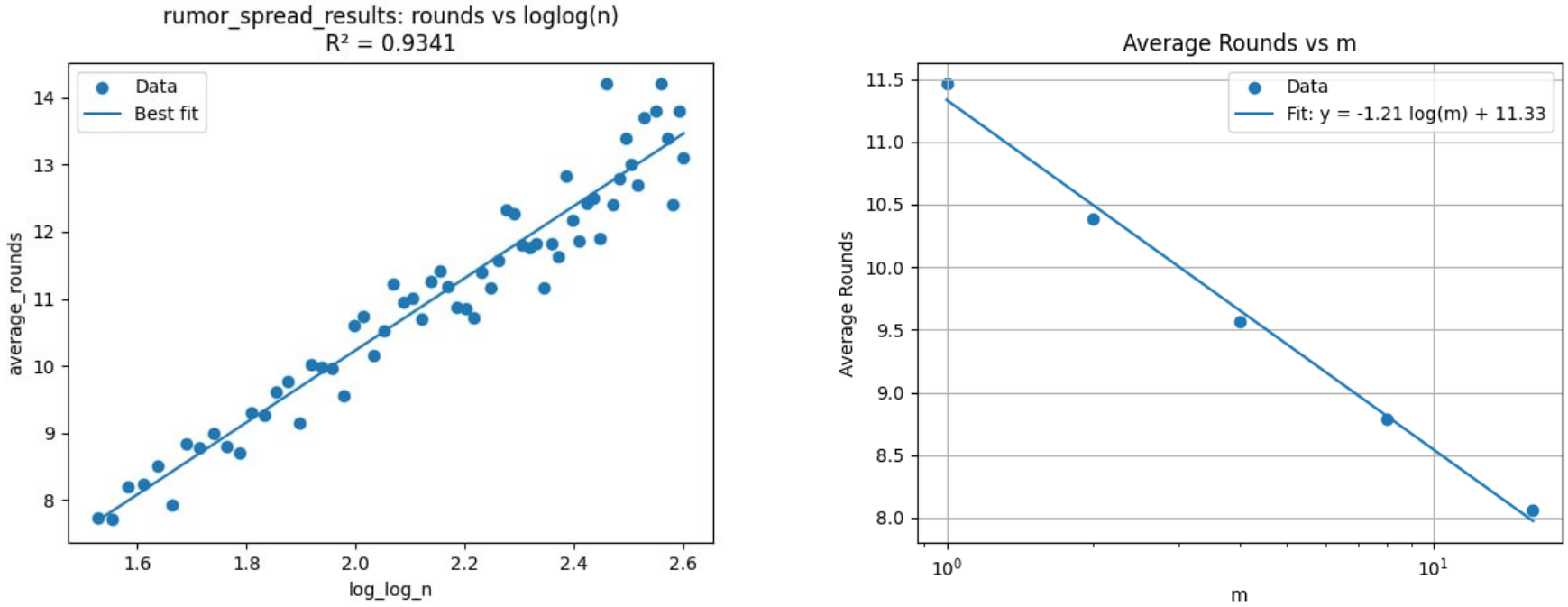


Figure: Left: Time vs total population; Right: Time vs number of influencers (for fixed $n = 10^4$)

- More influencers (m) $\rightarrow$ faster spreading
- Reduces effective depth of propagation
- Early phase is effectively skipped

Limitations and Improvements

Key Limitations

- **Uniform Sampling:** Nodes choose targets blindly $\rightarrow$ high collisions and wasted calls
- **Memoryless Behavior:** No learning from past interactions $\rightarrow$ repeated inefficiencies
- **No Global Awareness:** Nodes cannot detect phase (early vs late) $\rightarrow$ no adaptive strategy

Possible Improvements

- **Adaptive Sampling:** Bias towards likely useful contacts
- **Local Memory:** Track success/failure rates to guide decisions
- **Decentralized Switching:** Infer phase using local signals

Applications

- **Gossip in distributed systems:** fast update dissemination in databases or caches.
- **Social networks:** modeling viral spread of information or rumors.
- **Epidemic modeling:** mathematical dual to disease spread (SIR models).

Key Takeaways

- Simple push/pull protocols inform all nodes in $\Theta(\ln n)$ rounds .
- The process exhibits strong concentration (low variance around the mean).
- Adding failures, parallel calls changes constants but keeps logarithmic behavior.
- Addition of influencer nodes reduces spreading time drastically (to $O(\log \log n)$)
- Understanding this guides design of efficient gossip and epidemic algorithms.

Thank You / Questions

Thank you!

Questions?